On the Erdős-Straus Conjecture: Algebraic Analysis and Modular Decomposition

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Abstract

This document presents a rigorous analysis and structural decomposition of the Erdős-Straus conjecture. We lay out axiomatic definitions, review relevant recent literature, isolate key lemmas for specific congruence classes, and provide an architecture for autoformalization in a proof assistant like Lean 4.

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1 Axiomatic Definitions and Context

Definition 1.1. *For every integer $n \in \mathbb{N}$ with $n \geq 2$, the Erdős-Straus equation is defined as the Diophantine equation:*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

where $x, y, z \in \mathbb{Z}_{>0}$.

1.1 Contextual Literature Research

The problem, formulated by Paul Erdős and Ernst G. Straus in 1948, posits that the equation always has a positive integer solution for $n \geq 2$. Recent studies by authors such as Miguel Angel Lopez have classified solutions into types, such as Type A and Type B, by defining a complete congruence system. Furthermore, Philemon Urbain Mballa has explored an unexpected connection between the discrete zeta function and the Erdős-Straus conjecture through additive decomposition. This methodology shows strong analogies to the algebraic manipulations used in solving Pell-Fermat equations, where solutions are grouped by modular invariants and analyzed via infinite descents or structural classifications.

2 Proof Strategy and Lemma Isolation

The fundamental approach involves partitioning the integers n by their congruence class modulo an integer, historically 840, although general cases can be studied for primes.

Lemma 2.1. *For $n = 4k + 3$ with $k \in \mathbb{N}$, the equation admits a positive integer solution.*

Proof. Let $n = 4k+3$. We set $x = (n+1)/4 = (4k+4)/4 = k+1$. Because $n \geq 3$, $k \geq 0$, and thus $x \geq 1$, ensuring $x \in \mathbb{Z}_{>0}$. We evaluate the remaining difference:

$$\begin{aligned}
\frac{4}{n} - \frac{1}{x} &= \frac{4}{n} - \frac{1}{k+1} \\
&= \frac{4(k+1) - n}{n(k+1)} \\
&= \frac{4k+4-(4k+3)}{n(k+1)} \\
&= \frac{1}{n(k+1)}
\end{aligned}$$

The remaining fraction $\frac{1}{n(k+1)}$ must be split into two unit fractions $\frac{1}{y} + \frac{1}{z}$. We employ the standard Egyptian fraction splitting identity:

$$\frac{1}{A} = \frac{1}{A+1} + \frac{1}{A(A+1)}$$

Applying this to $A = n(k+1)$, we set: $y = n(k+1) + 1$ and $z = n(k+1)(n(k+1)+1)$

Since $n \geq 3$ and $k \geq 0$, both y and z are strictly positive integers. By substitution:

$$\begin{aligned}
\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= \frac{1}{k+1} + \frac{1}{n(k+1)+1} + \frac{1}{n(k+1)(n(k+1)+1)} \\
&= \frac{1}{k+1} + \frac{n(k+1)+1+1}{(n(k+1)+1)(n(k+1)(n(k+1)+1))}
\end{aligned}$$

Wait, the direct sum of the last two is exactly $\frac{1}{n(k+1)}$. Let us show it explicitly:

$$\begin{aligned}
\frac{1}{y} + \frac{1}{z} &= \frac{1}{n(k+1)+1} + \frac{1}{n(k+1)(n(k+1)+1)} \\
&= \frac{n(k+1)}{n(k+1)(n(k+1)+1)} + \frac{1}{n(k+1)(n(k+1)+1)} \\
&= \frac{n(k+1)+1}{n(k+1)(n(k+1)+1)} \\
&= \frac{1}{n(k+1)}
\end{aligned}$$

Adding $\frac{1}{x} = \frac{1}{k+1}$, we get:

$$\begin{aligned}
\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= \frac{1}{k+1} + \frac{1}{n(k+1)} \\
&= \frac{n}{n(k+1)} + \frac{1}{n(k+1)} \\
&= \frac{n+1}{n(k+1)} \\
&= \frac{4k+4}{n(k+1)} \\
&= \frac{4(k+1)}{n(k+1)} \\
&= \frac{4}{n}
\end{aligned}$$

This proves the lemma for all $n \equiv 3 \pmod 4$. □

3 Architecture for Autoformalization

The formalization of the Erdős-Straus conjecture requires structuring the statement and the search space decomposition in Lean 4.

```
import Mathlib.Data.Nat.Basic
import Mathlib.Tactic.Omega
import Mathlib.Tactic.Ring

-- Axiomatic definition of the Erdos-Straus property
def SatisfiesErdosStraus (n : Nat) : Prop :=
  Exists (fun x => Exists (fun y => Exists (fun z => x > 0 /\ y > 0
      /\ z > 0 /\ 4 * x * y * z = n * (y * z + x * z + x * y))))
```

```
-- Lemma for mod 4 = 3
lemma erdos_straus_mod_4_3 (k : Nat) : SatisfiesErdosStraus (4 * k
   + 3) := by
  sorry

-- Main Theorem
theorem erdos_straus_conjecture (n : Nat) (hn : n >= 2) :
   SatisfiesErdosStraus n := by
  sorry
```